Volume I · Chapter 2 — Foundations of Clinical Data (EHR; code sets; privacy) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-I-Chapter-2-context.md` for the AI interaction log.

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Describe **EHR structure** and its core data domains, and distinguish structured vs. unstructured data.
- Navigate and **map medical code sets** — ICD-10-CM/PCS, CPT/HCPCS, SNOMED CT, LOINC — to the right purpose.
- Explain **data privacy/regulation** (HIPAA, GDPR) and what makes clinical data identifiable.
- Apply basic **de-identification** and justify **synthetic-data** use for development.

Topics (theory) — 10 h:

#	Topic	h
2.1	The EHR: data domains (demographics, encounters, dx, procedures, meds, labs, vitals, notes); structured vs. unstructured	2
2.2	Diagnosis & procedure code sets:	2

#	Topic	h
	ICD-10-CM/PCS, CPT/HCPCS	
2.3	Clinical terminologies: SNOMED CT (concepts/hierarchies), LOINC (labs)	2
2.4	Mapping, crosswalks & value sets between terminologies	1
2.5	Privacy & regulation: HIPAA (PHI, Safe Harbor / Expert Determination), GDPR (special-category data, lawful basis)	2
2.6	De-identification, synthetic data & data-governance basics	1

Datasets/tools: synthetic EHR; public browsers (ICD-10, SNOMED CT browser, LOINC search); Pandas.

Assessment: quiz on code sets & privacy (**40%**); crosswalk deliverable (**30%**); de-identification lab rubric (**30%**).

Key decisions: which terminology fits which purpose; when data is/ isn't de-identified; **synthetic vs. real** data for development.

References: plan §13 → *Clinical data, code sets & privacy*.

Hours: Theory **10** + Lab **6** = **16**.